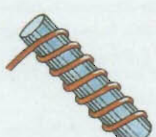
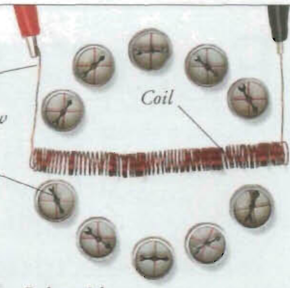
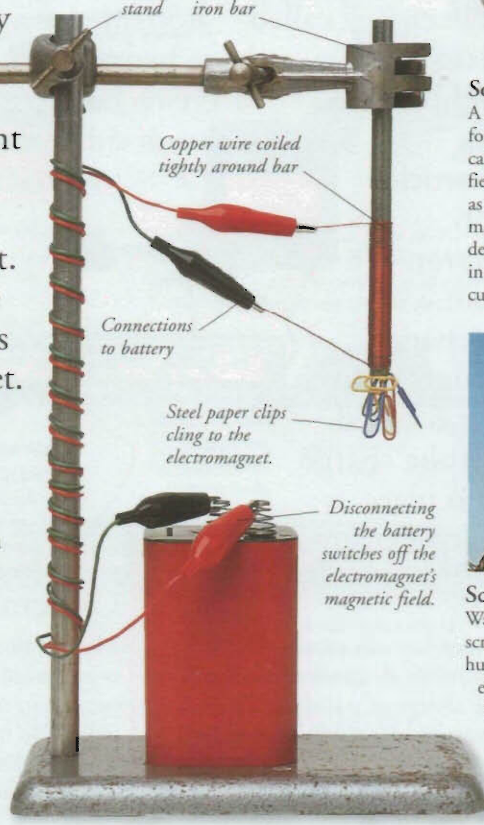
ELECTROMAGNETISM



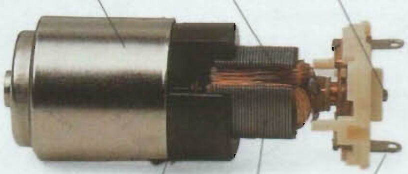
AT THE FLICK OF A SWITCH, an invisible force turns the drum of a washing machine 1,600 times every second. This force is called electromagnetism. It is a form of magnetism produced by electricity. When an electric current flows through a wire, it produces a magnetic field around the wire. Making the wire into a coil increases the strength of the magnetic effect. Winding the coil around an iron bar makes the magnetism even stronger. Any device that exerts electromagnetic forces is called an electromagnet.



Solenoid
A coil of current-carrying wire forms a type of electromagnet called a solenoid. The magnetic field around the coil is the same as that around an ordinary bar magnet. The field's strength depends on the number of turns in the coil and the amount of current flowing through the wire.



Scrapyard electromagnet
Waste metal is moved around a scrapyard by a crane carrying a huge electromagnet. When the electromagnet is switched on, it picks up metal scraps containing iron. The metal is moved to a different place and then dropped by switching off the electromagnet.



Electric motor

Inside an electric motor are wire coils surrounded by permanent magnets. Electricity flowing through the wire produces a magnetic field around each coil. The magnetism of the coils interacts with the magnetic fields of the permanent magnets. They push and pull on each other, making the coils rotate. This movement is used to drive machines such as electric drills.

Electric drill
An electric drill can quickly make a hole in wood, stone, and even some metals. Inside the body of the machine, gears harness the rotation of a powerful electric motor to drive the drill at high speed. A cooling fan prevents the drill from overheating.

Electromagnet

Most electromagnets consist of a coil of wire wrapped around an iron bar. When an electric current flows through the wire, a magnetic field forms around the electromagnet. The magnetism can be switched off by disconnecting the electricity supply.

Uses of electromagnetism

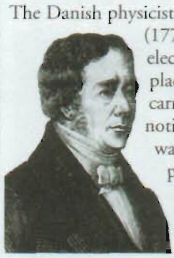
Some electrical appliances contain electric motors that use electromagnetism to produce movement. But electromagnetism is also used in many other ways, such as to make sound or detect hidden objects.

Loudspeaker
A loudspeaker contains a paper or plastic cone that vibrates and creates sound waves in the air around it. The cone is attached to a wire coil surrounded by a permanent magnet. The magnetic fields of the coil and the magnet interact. This causes the coil to move rapidly to and fro, making the cone vibrate.

Metal detector
Inside the walk-through arch of an airport metal detector are large coils of wire carrying an electric current. Any person who walks under the arch passes through the magnetic field produced by the coils. A hidden metal object will affect the strength of the field and trigger an alarm.

Transformer
Many electrical devices use a transformer to alter the voltage of an electrical supply. Inside a transformer are two wire coils. When a varying current flows through one coil, it produces a varying magnetic field. This field causes an electric current to flow through the second coil, but at a different voltage.

Hans Christian Oersted



The Danish physicist Hans Christian Oersted (1777–1851) discovered electromagnetism in 1820. He placed a compass near a wire carrying an electric current and noticed that the compass needle was deflected and no longer pointed north. Oersted realized that the current had produced a magnetic field around the wire.

Timeline

1799 Italian physicist Alessandro Volta invents the battery, which allows scientists to experiment with electric currents.

1820 Oersted's discovery of electromagnetism opens the way for the development of the electric motor and the electromagnet.

Faraday's electric motor

1821 English scientist Michael Faraday makes an electric motor, in which a current-carrying wire rotates around the pole of a magnet. It has no practical use.

Sturgeon's electromagnet

1828 English scientist William Sturgeon builds the first electromagnet – a coil of wire around an insulated iron bar.

1883 Croatian-born physicist Nikola Tesla invents the "induction motor" – the first practical motor.

1885 American engineer William Stanley invents the transformer.

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